Instance-Wise Domain Generalization for Cross-Scene Wetland Classification With Hyperspectral and LiDAR Data

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***Abstract*— Wetland is one of the three ecosystems in the world, and collaborative monitoring using hyperspectral images (HSIs) and light detection and ranging (LiDAR) has been important for wetland ecological protection. However, because of the domain shift of different images, cross-scene wetland classification of HSIs and LiDAR is a practical challenge, necessitating the development of models trained solely on the source domain (SD) and directly transferred to the target domain (TD) without retraining. To address this issue, an instance-wise domain generalization network (IDGnet) is proposed for HSI and LiDAR cross-scene wetland classification. An instance-wise random domain expansion module (IWR-DEM) is developed to simulate the domain shift, establishing the extended domain (ED). Specifically, the original HSI and LiDAR data are separated as semantic and background information in the frequency domain, a random background shift is applied to the HSI, and a semantic random shift is deployed to LiDAR. The HSI and LiDAR fusion features are extracted from the SD and ED by a weight-shared network. Multiple condition constraints are proposed for domain and class alignment, learning the domain-invariant and class-specific information and improving model generalization. Experiments conducted on two wetland datasets demonstrate the superiority of the proposed IDGnet for cross-scene wetland classification with HSI and LiDAR data. The codes will be available from the website: https://github.com/bigshot-g/IEEE_TGRS_IDGnet.**

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***Index Terms*— Cross-scene wetland classification, domain generalization, hyperspectral image (HSI), light detection and ranging (LiDAR).**

I. Introduction

WETLAND is one of the three ecosystems in the world, with significant environmental importance [1], [2]. With the rapid development of Earth observation technology, remote sensing has become a crucial method for wetland monitoring and protection [3], [4], [5], through which abundant data are acquired. However, due to the difficulty of verification and the high cost of manual interpretation, there is a limited number of labeled sample images, and cross-scene wetland classification, that is, the use of labeled images to classify unlabeled images, is a practical challenge.

A hyperspectral image (HSI) can capture a continuous and wide spectral range with a high-resolution spectral response for various objects and has found wide use in wetland classification [6], [7], [8]. In the task of cross-scene wetland classification, variations in collection conditions, such as changes in light, environment, and sensor conditions, can lead to spectral shifts between the source domain (SD) and the target domain (TD) in HSI data. In general, labeled images are in the SD, and unlabeled images are in the TD. It is well known that learning-based intelligent classifiers rely on the assumption that the training and test data belong to the same distribution. In practice, the phenomenon of spectral shift violates this assumption, which causes the poor generalization performance of traditional intelligent classifiers in cross-scene wetland classification.

With the development of transfer learning, domain adaption (DA) has shown promise in cross-scene HSI classification [9], [10], [11] and has been demonstrated to achieve distribution alignment across domains by minimizing the distribution discrepancy at the feature level and acquiring domain-invariant information. Current DA methods for cross-scene HSI classification are either feature alignment-based [12] or adversarial-based [13]. Those based on feature alignment employ metric criteria to minimize feature distribution differences, with the maximum mean discrepancy (MMD) criterion being extensively utilized. Zhang et al. [14] proposed a topological structure and semantic information transfer network (TSTnet), utilizing the MMD and graph optimal transmission for feature alignment of SD and TD. A subspace alignment

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and MMD cooperative method were developed to reduce the geometric and statistical shift [15]. Among other types of methods, Ning et al. [16] proposed a contrastive learning framework based on category matching and adopted instance-level discriminative feature embedding for feature alignment. Adversarial-based methods aim to reduce domain differences between SD and TD using a generative adversarial network (GAN), and maximum classifier discrepancy (MCD) has been a common method for DA [17], [18]. Huang et al. [19] combined a two-branch attention network and MCD, improving the ability of discriminative feature extraction and cross-scene classification. Fang et al. [20] developed a confident learning-based DA network (CLDA), which alternates training MCD and confident learning to reduce domain differences. An unsupervised joint adversarial domain adaptation method was proposed, which can reduce class- and domain-level feature differences between SD and TD by adversarial learning [21].

The above DA methods, despite having achieved satisfactory performance in cross-scene HSI classification, face two practical problems in cross-scene wetland classification. First, vegetation in wetlands consists primarily of mixed trees and shrubs, resulting in similar spectral characteristics across different types of vegetation. Therefore, to rely solely on spectral information is insufficient. Second, DA methods have access to TD test data during training, which may deviate from practical application. Because wetland monitoring must be timely, it is not practical to train on unlabeled data every time.

The above problems have been addressed often. Light detection and ranging (LiDAR) can provide the elevation information of an observed object, which is an effective complement to spectral information [22], [23]. Recently, a variety of methods have been proposed for HSI and LiDAR classification [24], [25], [26]. Zhao et al. [27] combined a hierarchical convolutional neural network (CNN) and Transformer to fuse multisource heterogeneous information, improving the performance of HSI and LiDAR joint classification. A multimodal fusion Transformer (MFT) network was proposed to fuse HSI and LiDAR, utilizing multihead cross-patch attention [28]. The combination of HSIs and LiDAR is effective in wetland classification, especially in distinguishing trees and shrubs [29]. Naturally, this has also been applied in cross-scene classification. Zhang et al. [11] developed a multilevel DA network to promote information collaboration between HSIs and LiDAR, with modality-related adapters and a mutual-aid classifier to aggregate discriminative information for improving cross-scene classification performance. Domain generalization (DG) exclusively relies on labeled SD data during training, which does not have access to unlabeled TD data, making its task setting more realistic compared to that of DA. In the past few years, DG has been widely adopted in computer vision [30], [31], [32], [33], [34]. Zhou et al. [35] proposed an instance-aware DG framework, which can obtain generalizable features by reducing the sensitivity of features to instance-specific styles. Chen et al. [36] developed a simulate–analyze–reduce learning paradigm for DG, which can build an auxiliary domain to learn the cause of domain shift. Therefore, DG-based methods also have been introduced in cross-scene HSI classification tasks [37], [38]. Zhang et al. [39] proposed a single-SD expansion network (SDEnet) for cross-scene HSI classification, utilizing an encoder–randomization–decoder architecture to generate the ED, and learning domain-invariant information through intraclass samples of SD and ED. The local joint feature was implicitly extended by style randomization, which can be used to learn domain-specific and domain-invariant features [40].

Existing research has only addressed one of the two aforementioned issues at a time. Combining HSI and LiDAR for cross-scene classification is still based on DA, and TD is still used in the training stage. DG-based methods only utilize HSIs and cannot deal with the phenomenon that different objects have similar spectra. To our knowledge, research on simultaneously solving the above two problems in the cross-scene wetland classification task is lacking. The improvement of wetland monitoring requires a multisource remote sensing data DG method for cross-scene wetland classification. However, data have different distribution characteristics, and it is a challenge to design a generalization network for multisource data while retaining the advantages of fusion features.

To address the above two issues, an instance-wise DG network, instance-wise domain generalization network (IDGnet), is proposed for cross-scene wetland classification. IDGnet implements domain extension for each instance of HSI or LiDAR and deploys multiple alignment constraints to improve generalization. IDGnet includes a domain expansion block, a feature fusion block, and a domain and class alignment block. To simulate the domain shift of different modalities, the discrete cosine transform (DCT) and a Gaussian filter are applied in the domain expansion block to separate semantic and background information from the original data in the frequency domain. Random background and semantic shifts are added to each instance of HSIs and LiDAR, respectively, to ensure effective ED representation. The feature fusion block utilizes the weight-shared network to extract the multisource features of SD and ED and fuses HSI and LiDAR features. In the domain and class alignment block, an instance narrowing constraint and supervised contrastive learning are deployed to obtain domain-invariant and class-specific representation. SD and ED are constrained by the prediction consensus strategy to further improve generalization ability. Real-world multisource wetland datasets collected in the Yellow River Delta coastal wetland are used for comparative experiments.

The main contributions of IDGnet are summarized as follows.

1) An instance-wise DG network is explored for cross-scene wetland classification. Different from existing work, IDGnet fuses HSIs and LiDAR for DG in the cross-scene wetland classification task, which is more significant in practical application than DA-based and single-source DG-based methods.
2) An instance-wise random domain expansion network is proposed to simulate the domain shift, which can apply background and semantic random shifts to HSIs and LiDAR, respectively, increasing the diversity of ED. In addition, instance-wise domain expansion ensures that the ED retains the characteristics of each instance and increases its reliability.
3) A strategy of multiple condition constraints is developed to enhance the generalization ability of multisource data. The instance narrowing constraint aims to pull the SD

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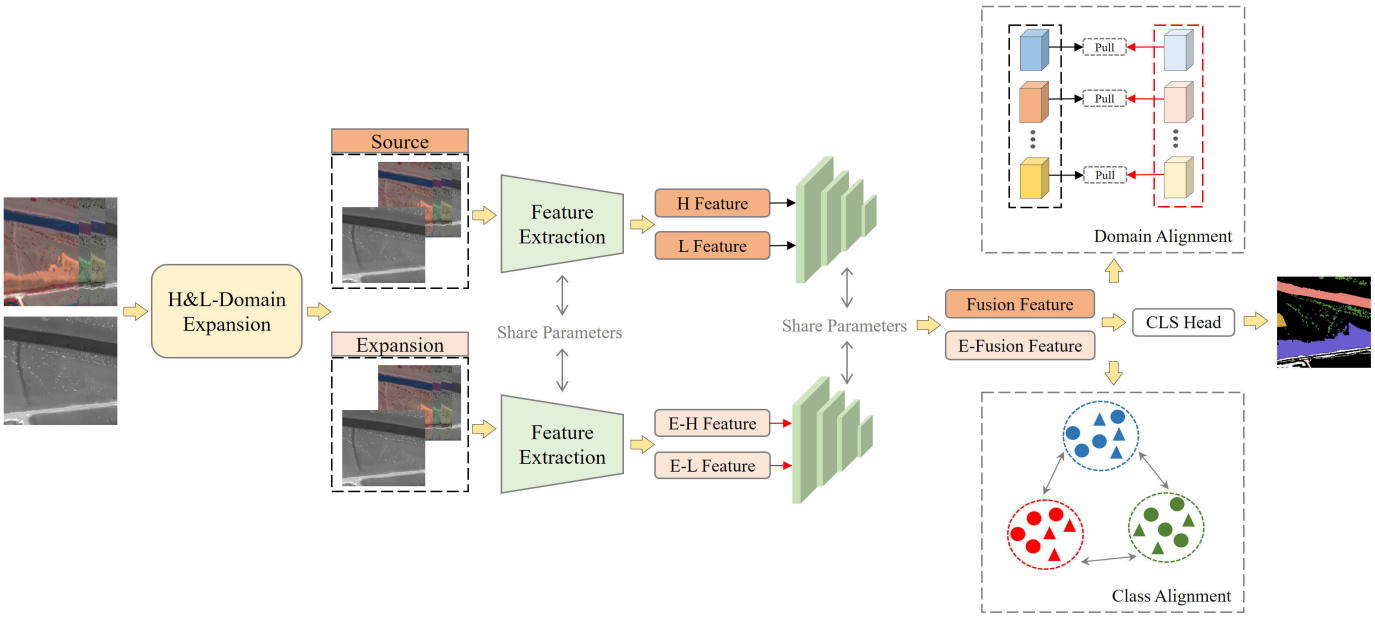


Fig. 1. IDGnet framework.

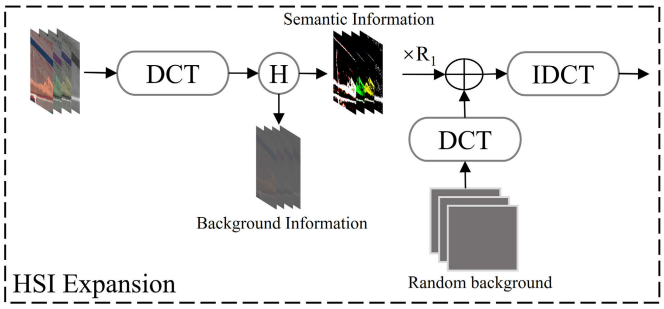


Fig. 2. HSI domain expansion.

and ED close and find the domain-invariant representation. Supervised contrastive learning can achieve the aggregation of the same classes in the feature space, obtaining the class-specific representation. The prediction consensus strategy is used to improve classification ability.

The rest of this article is organized as follows. Section II presents the details of the proposed IDGnet. Section III discusses experimental results and analyzes the effectiveness of IDGnet. We present our conclusions in Section IV.

II. Proposed Method

Here, we introduce the architecture and mathematics of IDGnet, whose framework is depicted in Fig. 1. IDGnet consists of: 1) an instance-wise random domain expansion module (IWR-DEM); 2) a multisource feature extraction and fusion module (MSF-EFM); and 3) a domain and class alignment module (DCAM).

A. Instance-Wise Random Domain Expansion Module

Compared with DA-based methods, DG-based methods do not access TD at training. Generalization cannot be achieved using only SD data, so the ED should be built to simulate the domain shift. We formulate the remote sensing image as

$$\boldsymbol{I}_{\mathrm{rs}} = f(\boldsymbol{Se} + \boldsymbol{B}) \tag{1}$$

where $\boldsymbol{I}_{\mathrm{rs}}$ represents a remote sensing image, such as HSIs or LiDAR. $\boldsymbol{Se}$ is the semantic information, $\boldsymbol{B}$ is the background information, and $f(\cdot)$ is a nonlinear transformation, representing the inherent properties of different domains.

It can be inferred from (1) that the domain shift is the reflection of the $\boldsymbol{Se}$ or $\boldsymbol{B}$ shift. Therefore, a random transformation is applied on $\boldsymbol{Se}$ or $\boldsymbol{B}$ to simulate the domain shift. The random transformation includes HSI domain expansion and LiDAR domain expansion. We note that there are two scales on the random transformation: batch level and instance level. At the batch level, all instances are treated as a whole, and batch-level random transformation will result in the homogenization of instance features, losing discrimination. Instance-level random transformation retains the characteristics of each instance and avoids interference from other instances. Hence, instance-level random transformation is adopted in IDGnet.

1) HSI Domain Expansion: Spectral information is the intrinsic information of the target object, and theoretically, the spectral information of the same object should be consistent. However, given the passive imaging mode of HSI remote sensing, the HSI is easily affected by changes in the environment, leading to the spectral change of objects. According to (1), the intrinsic spectral information of an object is considered to be the semantic information, and the imaging spectral change can be attributed to background information shift. Therefore, the random transformation of HSI domain expansion should be applied to background information $\boldsymbol{B}$. The background information shift is conducted in the frequency domain by a DCT. The low-frequency component of an image contains much background information, and the high-frequency component contains much semantic information [41]. So, as shown in Fig. 2, the DCT is first deployed to transform the input HSI from the time domain to the frequency domain. Then,

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a high-pass filter $H_h(u, v)$ is used to separate semantic and background information from an HSI in the frequency domain. A random fluctuation coefficient R_1 is then multiplied by the semantic information, and random background information is added. R_1 represents the spectral change caused by the light change, and the random background information represents changes in the imaging background caused by other conditions. The above process simulates the domain shift, and an ED is built. Finally, the inverse DCT (IDCT) is utilized to transform the ED to the time domain. HSI domain expansion can be formulated as

$$\boldsymbol{H}_e = \text{IDCT}((\text{DCT}(\boldsymbol{H}_s) \times H_h(u, v) \times R_1) + \boldsymbol{B}_{\text{ran}}) \tag{2}$$

where $\boldsymbol{H}_s$ and $\boldsymbol{H}_e$ are the respective SD and ED of HSIs. R_1 and $\boldsymbol{B}_{\text{ran}}$ are the respective random fluctuation coefficient and random background information. R_1 takes values from 0.6 to 1, and $\boldsymbol{B}_{\text{ran}} = \text{DCT}(b)$, where b represents spectral background reflection and ranges from 0 to 300. The Gaussian high-pass filter is defined as

$$H_h(u, v) = 1 - e^{-\frac{u^2+v^2}{2\lambda_h^2}} \tag{3}$$

with a cut-off frequency $\lambda_h = 110$.

2) LiDAR Domain Expansion: LiDAR has an active operation mode, which is not affected by the environment during work. The domain shift of LiDAR is caused more by the change in the detected object. The weather conditions of wetlands are complicated, and the vegetation can easily bend under the influence of wind, resulting in elevation changes. In addition, a particular kind of vegetation can have various elevations, due to the growth state. Therefore, according to (1), the domain shift of LiDAR can be attributed to semantic information shift. The background information of LiDAR is terrain information, which can be regarded as constant. The framework of LiDAR domain expansion is shown in Fig. 3. Similar to HSI domain extension, the DCT is first deployed to transform the input LiDAR data from the time domain to the frequency domain, and a high-pass filter $H_l(u, v)$ is used to separate semantic information and background information from LiDAR data. Then, a random fluctuation coefficient R_2 is multiplied by the semantic information to simulate elevation changes. Different from HSI domain expansion, the background information B obtained from $H_l(u, v)$ is added to the new semantic information, which means adding the terrain information. Finally, IDCT transforms the ED to the time domain. LiDAR domain expansion can be calculated as

$$\boldsymbol{L}_e = \text{IDCT}((\text{DCT}(\boldsymbol{L}_s) \times H_l(u, v) \times R_2) + \boldsymbol{B}) \tag{4}$$

where $\boldsymbol{L}_s$ and $\boldsymbol{L}_e$ are the SD and ED, respectively, of LiDAR; R_2 is the random fluctuation coefficient, which ranges from 0.7 to 1; B is the background information, defined as $\boldsymbol{B} = \text{DCT}(\boldsymbol{L}_s) \times (1 - H_l(u, v))$; and $H_l(u, v)$ is the Gaussian high-pass filter

$$H_l(u, v) = 1 - e^{-\frac{u^2+v^2}{2\lambda_l^2}} \tag{5}$$

where λ_l denotes the cut-off frequency and $\lambda_l = 80$.

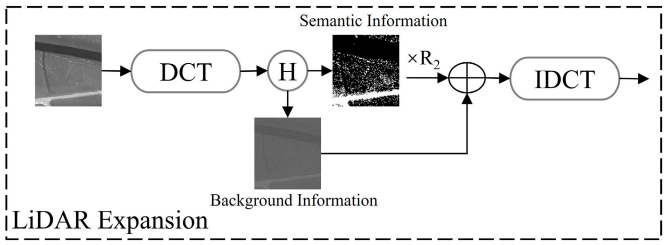


Fig. 3. LiDAR domain expansion.

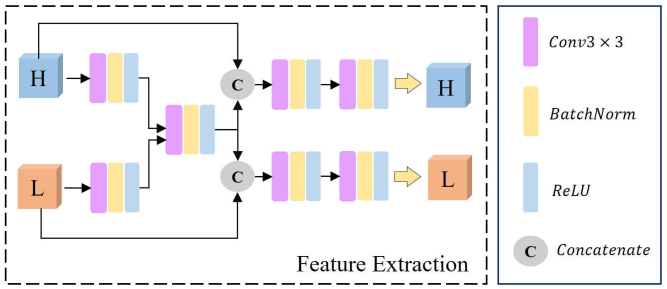


Fig. 4. Framework of feature extraction.

B. Multisource Feature Extraction and Fusion Module

After the IWR-DEM, the SD and ED are sent to the MSF-EFM to obtain the HSI-LiDAR fusion features. To avoid bias and noise introduced during data processing, weight-shared feature extraction and fusion networks are developed to ensure that both SD and ED are processed by a unified network. As shown in Fig. 4, the feature extraction network is composed of multiple convolutional (Conv) layer-batch normalization (BN) layer-ReLU activation layer blocks with different functions. First, two Conv-BN-ReLU blocks, ϕ_{he} and ϕ_{le}, are utilized to embed the HSI and LiDAR into different feature spaces, which have the same dimension. Then, due to the consistent observation targets of HSIs and LiDAR, a weight-shared Conv-BN-ReLU block ϕ_p is used to extract the public geospatial information. Third, to retain the spectral and elevation features, a residual structure is applied to enhance the specific properties of HSIs and LiDAR. Finally, two independent aggregation branches, ϕ_{ha} and ϕ_{la}, are adopted to aggregate the geospatial information and specific properties, acquiring the HSI and LiDAR features. The feature extraction stage can be formulated as

$$\begin{aligned} \boldsymbol{F}_h &= \phi_{ha}(\phi_p(\phi_{he}(\boldsymbol{H})) + \boldsymbol{H}) \\ \boldsymbol{F}_l &= \phi_{la}(\phi_p(\phi_{le}(\boldsymbol{L})) + \boldsymbol{L}) \end{aligned} \tag{6}$$

where $\boldsymbol{H}$ and $\boldsymbol{L}$ are the respective input data of HSIs and LiDAR, and $\boldsymbol{F}_h$ and $\boldsymbol{F}_l$ are the corresponding output features.

The feature fusion network is a traditional three-layer convolutional block, aiming to fuse $\boldsymbol{F}_h$ and $\boldsymbol{F}_l$. For SD and ED, their feature extraction and fusion processes are independent, while the weight is shared. The advantage of MSF-EFM is its ease of deployment and efficient execution. Compared with other feature fusion networks, MSF-EFM only extracts the principal component features of multisource data, while fine-grained feature compensation is realized by multiple condition constraints.

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C. Domain and Class Alignment Module

Let us assume the output features of MSF-EFM are $\boldsymbol{F}_s$ and $\boldsymbol{F}_e$, which represent the respective multisource fusion features of SD and ED, respectively. A feature representation employed for a classification task generally consists of four kinds of information: domain-invariant, domain-specific, class-specific, and class-invariant. Domain-specific information has a special property caused by a domain shift. SD and ED have different domain-specific information. Class-invariant information is that of interclass interference, which represents a common attribute between classes. Domain-specific and class-invariant information are negative factors for cross-scene classification. Therefore, the goal of DCAM is to extract domain-invariant and class-specific information from fusion features $\boldsymbol{F}_s$ and $\boldsymbol{F}_e$ and reduce the domain-specific and class-invariant information.

To achieve the above goal, a strategy of multiple condition constraints is developed to obtain the domain-invariant and class-specific information. Existing methods typically use adversarial learning to acquire the domain-invariant information; these methods are highly sensitive to hyperparameters and are difficult to train. In DCAM, the instance narrowing constraint is adopted to extract the domain-invariant information of SD and ED. This constraint aims to build a common representation between instances of SD and ED. By pulling the instances in $\boldsymbol{F}_s$ and $\boldsymbol{F}_e$ close together, instances share the same semantic content while perceiving the representation of another domain. In this way, $\boldsymbol{F}_s$ and $\boldsymbol{F}_e$ tend to have a similar intermediate representation, which indicates that the fusion features have learned domain-invariant information. The instance narrowing constraint can be formulated as

$$L_{\text{pull}} = \sqrt{\frac{1}{B}\sum_{i=1}^{B}\left(\boldsymbol{F}_s^i - \boldsymbol{F}_e^i\right)^2} \tag{7}$$

where B is the batch size. A batch contains both SD and ED data, and $\boldsymbol{F}_s^i$ and $\boldsymbol{F}_e^i$ are the ith instance of SD and ED, respectively, in one batch.

To learn the class-specific information, supervised contrastive learning is applied to align class information, and the prediction consensus strategy is utilized to ensure that both SD and ED have the same label in each corresponding instance. Hence, instances with the same label are set as positive samples, and those with different labels as negative samples. All instances in SD and ED are mixed, and only category information is considered, ignoring domain differences. We push instances with different labels away from each other and pull those with the same labels close. In this process, the interclass distance of $\boldsymbol{F}_s$ and $\boldsymbol{F}_e$ is increased, while the intraclass distance is decreased. As a result, the fusion feature becomes more robust and discriminative, promoting the learning of class-specific information in networks. Supervised contrastive learning can be formulated as

$$L_{\text{con}} = \frac{1}{2B}\sum_{i=0}^{2B}\frac{-1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a\in A(i)}\exp(\text{sim}(z_i, z_a)/\tau)} \tag{8}$$

where $2B$ is the number of instances in SD and ED. $P(i)$ and $A(i)$ are the positive and negative instance sets, respectively. $|P(i)|$ is the number of positive samples. z_i is the feature embedding of an instance, and z_p and z_a are its respective positive and negative samples. $\text{sim}(\cdot)$ is cosine similarity. τ is a temperature hyperparameter that controls the concentration level, which is set to 0.07.

After the DCAM, the output feature maps are sent to a classification block to obtain the final classification map.

D. Loss Function

We propose the loss function of IDGnet when optimizing the parameters. Besides the loss of the instance narrowing constraint and supervised contrastive learning, the overall loss function includes classification loss. According to the prediction consensus strategy, the classification results for SD and ED should be the same. Therefore, the classification loss can be calculated as

$$L_{\text{cs}} = -\left(\sum_{i=1}^{B} y_i \cdot \log\left(p_s^i\right) + \sum_{i=1}^{B} y_i \cdot \log\left(p_e^i\right)\right) \tag{9}$$

where y_i is the one-hot embedding of the ground-truth label, which is the required output; and p_s^i and p_e^i are the respective prediction probability distributions of SD and ED, which are obtained from the output of IDGnet.

Finally, the overall loss function is formulated as

$$L_{\text{all}} = L_{\text{cs}} + \lambda_1 L_{\text{pull}} + \lambda_2 L_{\text{con}} \tag{10}$$

where λ_1 and λ_2 are balance coefficients to control the relative weights of the loss terms.

III. Experimental Results and Analysis

All experiments were performed on a platform equipped with an NVIDIA GeForce RTX 3090 GPU (24 GB GDDR6X) and an Intel Core i9-10900K CPU, with 96 GB of RAM. All programs were built using PyTorch 1.10.0 and Python 3.7.11. Experiments employed the Adam algorithm to optimize all trainable parameters in IDGnet. SD data were only employed for IDGnet training, while TD data were exclusively utilized for testing. In the training phase, we set the number of epochs to 70, the batch size to 16, and the learning rate to 0.00001. In the testing phase, we utilized the metrics of overall accuracy (OA), average accuracy (AA), and the kappa coefficient (K) to evaluate classification performance.

A. Data Description

All the datasets were collected in the Yellow River Delta National Nature Reserve, Shandong, China. The HSI data were collected by a Cubert Utilities FIREFLEYE S185 frame camera mounted on a DJI M600 Pro six-rotor UAV platform. The HSI data contain 126 spectral bands with a spectral resolution of 4 nm, ranging from 450 to 950 nm. The LiDAR data were obtained by a D-LiDAR 2000 camera on a FEIMA D2000s four-rotor UAV platform. The spatial resolution of HSIs and LiDAR was 0.5 m. The data acquired from UAV should be preprocessed before use. First, the original HSI data were conducted the radiometric calibration and geometric correction. Second, the original point clouds were conducted the point resolving, strip adjustment, and raster conversion to get the LiDAR image of the flight area. Next, the HSI

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TABLE I
INFORMATION OF SD AND TD ON NW-1 AND NW-2

No.	color	Class	NW-1 (Source)	NW-2 (Target)
1		Salix matsudana	20	55495
2		Populus	20	14378
3		Tamarix chinensis	20	49129
4		Yellow River	20	47444
5		Highway	20	6798
6		Phragmites australis	20	91216
		Total	120	264460

TABLE II
INFORMATION OF SD AND TD ON NW-1 AND NW-3

No.	color	Class	NW-1 (Source)	NW-3 (Target)
1		Salix matsudana	20	33135
2		Tamarix chinensis	20	4944
3		Highway	20	12194
4		Phragmites australis	20	108334
		Total	80	158607

and LiDAR data were projected into the projected coordinate system WGS 1984 UTM Zone 50 N, aligned pixel by pixel to acquire the multisource dataset. Finally, combined with field data and UAV-based RGB images, the multisource remote sensing data were interpreted to get wetland classification datasets.

1) Natural Willow Datasets: The datasets consist of three sets of multisource remote sensing data, that is, NW-1, NW-2, and NW-3, each composed of an HSI- and LiDAR-derived digital surface model (DSM). The Natural Willow datasets were acquired in June 2023. The dominant vegetation species is Salix matsudana, which is widely distributed in the Yellow River estuary and is important terrestrial vegetation in the Yellow River Delta coastal wetlands. NW-1, NW-2, and NW-3 correspond to three distinct regions, with respective sizes of 1197 × 1445, 935 × 2060, and 1073 × 1329. Focusing on the cross-area analysis over the same temporal, NW-1 was selected as the source data, and NW-2 and NW-3 were the target data. Tables I and II show the number of samples of source and TD.

2) Tamarisk Datasets: The Tamarisk datasets, Tamarisk-S and Tamarisk-AUT, contain multisource remote sensing data, each including HSI and LiDAR data. Tamarisk-S and Tamarisk-AUT were collected in the same area (the dominant vegetation species is Tamarisk) in July and September 2022, respectively. Tamarix chinensis is typical vegetation in the Yellow River Delta coastal wetland. The respective sizes of Tamarisk-S and Tamarisk-AUT are 779 × 1346 and 784 × 1204. Focusing on cross-season analysis over the same area, Tamarisk-S was selected as the source data and Tamarisk-AUT as the target data. The number of samples of source and TD is listed in Table III.

The SD samples for three DG tasks are randomly selected from the SD dataset. All samples of the TD dataset are TD samples.

TABLE III
INFORMATION OF SD AND TD ON TAMARISK-S AND TAMARISK-AUT

No.	color	Class	Tamarisk-S (Source)	Tamarisk-AUT (Target)
1		Desiccated Spartina alterniflora	20	52284
2		Phragmites australis	20	9660
3		Tidal creek	20	47660
4		Suaeda salsa	20	24305
5		Tamarix chinensis	20	33544
6		Roads	20	2691
		Total	120	170144

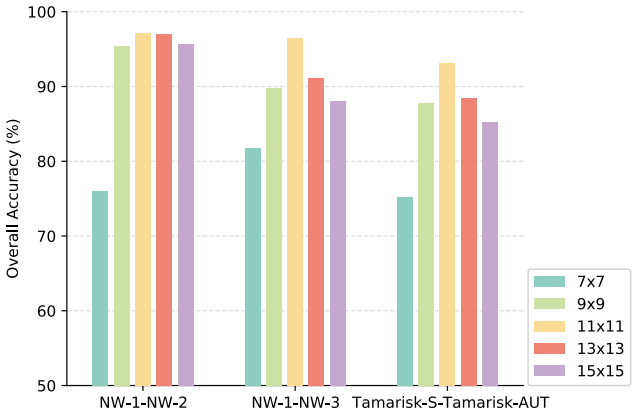


Fig. 5. Classification accuracy on different patch sizes.

B. Sensitivity Analysis

Sensitivity analysis explores the influence of model parameters, whose optimal values we seek for IDGnet to achieve the best performance in cross-scene wetland classification. We evaluated the effect of the patch size and the balance coefficients of the overall loss function in three DG tasks.

1) Effect of Different Patch Sizes: The patch of input data is composed of the central pixel and its neighborhood, and its size affects the amount of information carried by the patch. Fig. 5 shows the effect of patch size on cross-scene classification accuracy. We used patch sizes of 7 × 7, 9 × 9, 11 × 11, 13 × 13, and 15 × 15 in experiments. It can be found that as the patch size increases, the classification accuracy presents a trend of first increasing and then decreasing. When patch sizes increase from 7 × 7 to 11 × 11, the classification accuracy of IDGnet consistently improves. This is because large patches have more abundant spectral, elevation, and spatial information. However, classification performance declines as the patch size increases from 11 × 11 to 15 × 15. This phenomenon is manifested to different extents in three DG tasks. This is because overly large patches introduce redundancy and interference information. The spatial resolution of the input data is 0.5 m, resulting in large patches (i.e., 13 × 13, 15 × 15) with a coverage area of 6.5–7.5 m, which exceeds the crown breadth of most wetland vegetation, introducing interference information of other vegetation. According to the results shown in Fig. 5, the patch size can affect cross-scene classification performance. To obtain the best performance in the DG task, the patch size of input data was set to 11 × 11.

2) Effect of Hyperparameters on Loss Function: The hyperparameters λ_1 and λ_2 control the weights of L_{pull} and L_{con},

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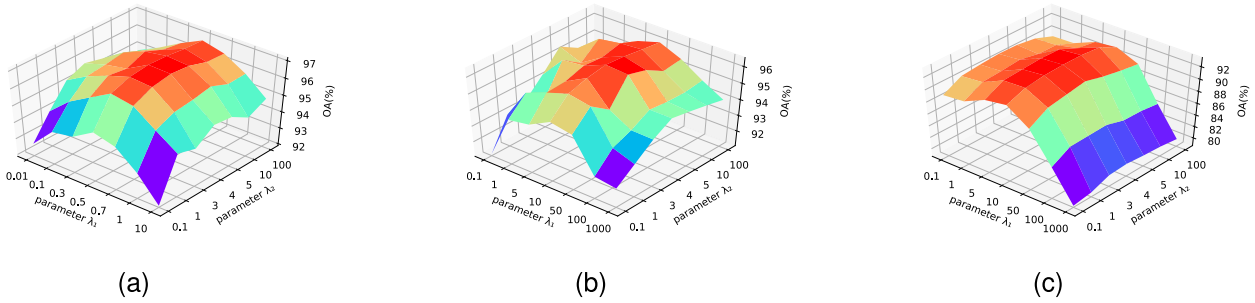


Fig. 6. Classification accuracy on different hyperparameters in the loss function. (a) NW-1-NW-2 task. (b) NW-1-NW-3 task. (c) Tamarisk-S-Tamarisk-AUT task.

TABLE IV
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON DOMAIN EXPANSION

	NW-1-NW-2			NW-1-NW-3			Tamarisk-S-Tamarisk-AUT		
	OA (%)	AA (%)	K (×100)	OA (%)	AA (%)	K (×100)	OA (%)	AA (%)	K (×100)
Without domain expansion	93.93	86.39	92.01	94.25	94.71	88.84	89.95	86.07	86.81
Only LiDAR domain expansion	94.34	90.37	92.62	94.81	94.74	89.83	91.22	88.23	88.50
Only HSI domain expansion	95.83	95.41	94.58	95.38	96.76	90.96	92.41	90.11	90.09
With domain expansion	97.07	96.25	96.18	96.42	96.37	92.89	93.15	91.18	91.05

respectively, in the loss function. Appropriate hyperparameters can improve the performance of domain and class alignment and enhance the cross-scene wetland classification accuracy. To evaluate the influence of the hyperparameters λ_1 and λ_2, Fig. 6 reports the results of quantitative comparisons under grid search.

As shown in Fig. 6, the classification accuracy exhibits a gradient, with lower values at the periphery and higher values toward the center. This shows that classification performance in the DG task is degraded when λ_1 and λ_2 are too high or low. The reasons can be summarized as follows. When the hyperparameters are small, the alignment effect is not obvious, and it is difficult to align the domain and class into an implicit representation with domain-invariant and class-specific information. When the hyperparameters are too large, it is easy to make the model fix either SD or ED and ensure that the unfixed domain does not approach the fixed one, resulting in a learned implicit representation containing only SD or ED information. When hyperparameters are too large in L_{pull} and L_{con}, the network pays less attention to classification loss and reduces classification accuracy.

The above phenomenon shows that classification performance is sensitive to λ_1 and λ_2. It should be noted that the optimal range of hyperparameters varies across different DG tasks. From Fig. 6, λ_2 exhibits the same values and trends in three DG tasks, while λ_1 behaves differently. From these results, it can be inferred that the best range of λ_1 is 0.5–5, and the best value of λ_2 is 2.

C. Ablation Analysis

Ablation studies were used to evaluate the contribution of vital components in the proposed IDGnet. We conducted ablation analyses of IWR-DEM, MSF-EFM, and DCAM on three DG tasks.

1) Ablation Analysis of Domain Expansion: The ablation experiment of IWR-DEM explored the effectiveness of domain expansion. To evaluate the effect of HSI and LiDAR domain expansion, three ablation experiments were conducted, with classification results as listed in Table IV. It can be found that "With domain expansion" obtains OA values of 97.07%, 96.42%, and 93.15% in three DG tasks, which provides 1%–4% improvement over other settings. This phenomenon shows the effectiveness of the proposed domain extension method in cross-scene classification.

Although the classification accuracy of "Without domain expansion" is poor, an average OA of over 90% is obtained in three DG tasks. This can be attributed to the class alignment module. The domain alignment module is invalid in the absence of domain expansion, but the class alignment module still extracts the class-specific information, which enhances the classification performance of no domain expansion model. Even so, the AA of "Without domain expansion" is unsatisfactory, especially in the NW-1-NW-2 task, which is about 10% below that of the best performance, further demonstrating the advantage of domain expansion.

Using only LiDAR or HSI domain expansion also results in an approximately 1%–2% reduction in classification performance, which shows the effectiveness of multi-SD generalization. Compared with the classification results of "Only LiDAR domain expansion" and "Only HSI domain expansion," "Only HSI domain expansion" is reliably a stable 1.5% higher than "Only LiDAR domain expansion" in terms of OA, and a similar phenomenon also appears in AA and K. This indicates that spectral information is more susceptible to domain shift, which is consistent with the passive mode of HSI sensors.

According to the above analysis, IWR-DEM effectively simulates domain shift and successfully improves the performance of cross-scene wetland classification in DG tasks.

2) Ablation Analysis of Feature Extraction and Fusion: MSF-EFM was utilized to embed and fuse multisource remote sensing images, that is, HSIs and LiDAR. An ablation experiment was conducted to verify the contribution of MSF-EFM, with classification results as reported in Table V. It can be seen that "With MSF-EFM" achieves better performance than "Without MSF-EFM" on three DG tasks. Taking the

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TABLE V
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON FEATURE EXTRACTION AND FUSION

	NW-1-NW-2			NW-1-NW-3			Tamarisk-S-Tamarisk-AUT		
	OA (%)	AA (%)	K ((×100)	OA (%)	AA (%)	K ((×100)	OA (%)	AA (%)	K ((×100)
Without MSF-EFM	86.80	85.55	83.03	91.07	94.19	83.38	87.35	86.81	83.53
With MSF-EFM	97.07	96.25	96.18	96.42	96.37	92.89	93.15	91.18	91.05

TABLE VI
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON MULTIPLE CONDITION CONSTRAINTS STRATEGY

	NW-1-NW-2			NW-1-NW-3			Tamarisk-S-Tamarisk-AUT		
	OA (%)	AA (%)	K (×100)	OA (%)	AA (%)	K (×100)	OA (%)	AA (%)	K (×100)
Without alignment	91.96	84.55	89.49	89.49	93.39	80.77	90.66	88.61	87.83
Only domain alignment (L_{pull})	93.89	85.48	91.98	91.59	94.47	84.23	90.98	89.35	88.25
Only class alignment (L_{con})	95.40	93.70	93.97	94.24	94.65	88.81	92.05	90.17	89.63
With all alignment ($L_{pull} + L_{con}$)	97.07	96.25	96.18	96.42	96.37	92.89	93.15	91.18	91.05

TABLE VII
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON NW-1-NW-2

Class	MFT		HCT		MCD		CLDA		TSTnet		SDEnet		LLURnet		IDGnet	
	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L
Salix matsudana	74.16	94.61	86.70	93.80	44.42	79.59	44.79	48.70	71.91	79.23	88.23	63.48	62.21	**98.87**	97.99	94.46
Populus	0.97	21.21	0	22.21	71.94	70.84	92.59	**97.02**	8.39	77.72	0	71.83	17.14	0	24.04	87.76
Tamarix chinensis	7.92	68.12	30.59	76.06	24.72	89.55	16.75	**99.79**	71.30	90.76	80.62	90.72	62.81	86.89	0	96.84
Yellow River	28.19	3.77	98.60	99.99	100.0	99.94	77.68	100.0	99.97	99.88	100.0	99.55	100.0	99.73	100.0	**100.0**
Highway	53.69	99.81	90.26	99.87	99.94	**100.0**	99.66	97.40	98.93	99.75	99.21	99.99	71.59	99.57	99.96	99.96
Phragmites australis	67.12	78.68	38.40	75.75	25.79	94.40	44.55	91.10	69.50	**98.92**	1.60	97.79	2.97	84.22	47.51	98.50
OA (%)	46.68±2.51	64.04±3.01	57.13±2.78	81.65±2.43	47.23±2.20	90.25±3.43	49.41±2.52	85.90±2.58	73.24±2.50	92.31±2.30	54.54±1.66	88.24±1.78	46.46±1.47	86.39±1.50	58.76±1.96	**97.07±1.08**
AA (%)	38.68±1.87	61.03±2.66	57.43±2.63	77.95±2.51	61.14±1.69	89.06±2.74	62.67±2.08	89.00±2.47	70.00±1.87	91.04±2.66	61.61±2.33	87.22±1.18	52.79 ±1.88	78.22±1.84	61.58±1.51	**96.25±1.86**
K (×100)	30.71±2.56	54.84±3.21	45.59±2.91	76.2±3.17	34.25±2.27	87.34±3.86	36.68±3.77	82.05±2.63	65.03±2.54	90.01±3.53	43.26±2.85	84.81±2.52	34.74±1.63	82.27±2.01	47.36±2.03	**96.18±1.23**

NW-1-NW-2 task as an example, "With MSF-EFM" achieves a higher OA, AA, and K by 10.2%, 10.7%, and 13.1%, respectively, compared to "Without MSF-EFM."

SD and ED utilize the weight-shared feature extraction–fusion network, which does not introduce noise due to network initialization and the normalization deviation caused by BN layers. In the feature extraction stage, the public geospatial information and enhanced specific properties obtained from feature interaction and the residual structure improve the discriminability of HSI and LiDAR features. Accordingly, MSF-EFM is effective for final classification.

3) Ablation Analysis of Domain and Class Alignment: To evaluate the effectiveness of domain and class alignment, we performed ablation experiments on the proposed IDGnet, with the conditions of being without alignment, using only domain alignment, and using only class alignment. From the results in Table VI, obvious performance degradation can be seen when IDGnet is deployed without alignment. Taking the NW-1-NW-3 task as an instance, the OA of "Without alignment" is only 89.49%, which is the worst performance in the experiments. The AA of "Only domain alignment" and "Only class alignment" are, respectively, about 1.1% and 1.3% higher than "Without alignment." "With all alignment" obtains the best classification performance, with an OA about 7%, 5%, and 5% higher than "Without alignment," "Only domain alignment," and "Only class alignment," respectively. It is clear that only domain alignment or class alignment can fully capture domain-invariant and class-specific information. The results verify the influence of DCAM in improving the performance of cross-scene wetlands classification in the DG task.

Conversely, the OA, AA, and K values of "Only class alignment" are always higher than those of "Only domain alignment" in three DG tasks. This demonstrates that class-specific information has more influence on classification accuracy than domain-invariant information in the cross-scene wetland classification task, which is determined by downstream tasks. This result confirms the conclusion above that even without a domain expansion model, class-specific information can enhance classification performance.

D. Cross-Scene Classification Performance

The cross-scene classification performance of the proposed IDGnet was compared to that of several other methods, including two HSI-LiDAR classification models, three DA-based methods, and two DG-based methods. The HSI-LiDAR methods are the MFT network [28] and hierarchical CNN with Transformer (HCT) [27]. DA-based methods are the MCD method [17], CLDA [20], and TSTnet [14]. DG-based methods are the single-source domain expansion network (SDEnet) [39] and locally linear unbiased randomization network (LLURnet) [40]. For a fairer comparison, the HSI and LiDAR are concatenated into a cube as the input of MCD, CLDA, TSTnet, SDEnet, and LLURnet. All methods were implemented following the configurations in the corresponding published papers, and parameters were optimized. To verify the effectiveness of multisource remote sensing data classification, we also conducted experiments using only HSI data. Tables VII–IX report the classification results of different methods in three DG tasks. Experiments were performed ten times, and the average values of metrics were determined. IDGnet was shown to be superior to other methods.

According to Tables VII–IX, HSI-LiDAR shows a significant improvement in classification performance compared with only using HSIs, which shows the effectiveness of using both HSIs and LiDAR for wetland classification. After fusing LiDAR data, it is noteworthy that Natural Willow datasets show a significantly greater increase in classification accuracy than Tamarisk datasets. This is because the vegetation in

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TABLE VIII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON NW-1-NW-3

Class	MFT		HCT		MCD		CLDA		TSTnet		SDEnet		LLURnet		IDGnet	
	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L
Salix matsudana	68.42	89.04	98.60	99.19	95.88	99.95	84.12	73.93	19.77	92.09	25.99	93.65	0	**99.99**	99.83	99.33
Tamarix chinensis	57.83	82.97	22.82	47.29	58.19	54.25	82.22	96.93	70.47	95.33	10.07	**99.51**	2.57	1.05	2.06	90.78
Highway	100.0	99.96	99.84	99.97	99.98	98.87	100.0	99.77	99.90	98.91	100.0	94.63	100.0	86.56	**100.0**	99.99
Phragmites australis	50.69	74.06	10.94	46.72	60.47	95.30	64.33	95.69	64.60	**96.44**	36.79	85.49	76.57	84.96	36.76	95.39
OA (%)	58.40±4.11	79.46±2.04	36.46±3.66	61.80±2.27	70.83±3.15	95.27±2.24	71.76±2.46	91.49±2.09	58.13±1.86	95.69±1.65	38.56±1.93	88.34±1.82	60.07±2.56	85.61±2.11	53.71±1.48	**96.42±0.96**
AA (%)	69.23±2.66	86.51±1.70	58.05±4.01	73.29±3.21	78.63±3.35	87.09±2.35	82.67±2.82	91.58±1.53	63.69±2.19	95.69±1.92	43.21±2.14	93.32±1.88	44.78±1.87	68.14±1.49	59.66±1.89	**96.37±1.23**
K (×100)	38.70 ±4.62	64.50±3.65	20.30±3.93	44.72±3.69	54.61±4.40	90.53±2.40	55.32±3.55	83.37±3.11	32.24±2.68	91.38±2.23	6.78±2.49	78.64±2.43	16.15±2.85	73.05±2.75	32.42±2.11	**92.89±1.45**

TABLE IX

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON TAMARISK-S-TAMARISK-AUT

Class	MFT		HCT		MCD		CLDA		TSTnet		SDEnet		LLURnet		IDGnet	
	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L	H	H+L
Desiccated Spartina alterniflora	80.65	92.49	84.05	86.68	94.72	95.90	97.12	96.74	98.25	93.98	79.23	87.33	95.95	84.91	**98.25**	96.35
Phragmites australis	69.00	54.39	43.06	61.76	79.23	73.76	78.32	**90.58**	69.64	74.49	66.17	80.94	50.38	62.89	68.52	79.91
Tidal creek	98.92	93.63	99.71	**99.80**	99.19	98.68	99.80	99.56	97.96	98.64	99.53	99.53	99.12	99.78	99.74	99.59
Suaeda salsa	2.82	2.92	30.68	79.26	38.42	63.36	44.37	65.06	55.59	85.02	49.15	84.63	0.88	66.12	54.01	**83.37**
Tamarix chinensis	60.04	91.94	82.04	54.13	98.17	96.37	95.56	90.02	81.05	89.86	75.90	73.81	61.30	79.28	**96.85**	89.48
Roads	**99.57**	94.54	99.48	98.25	93.53	85.06	98.51	96.99	95.69	74.73	93.87	82.46	96.77	61.95	96.99	98.40
OA (%)	74.05±2.23	77.77±2.58	78.33±2.19	81.65±2.59	87.71±2.69	90.69±2.08	88.98±2.68	91.45±1.96	87.02±2.00	91.78±1.88	79.46±1.86	87.26±2.02	73.85±2.55	83.67±2.40	90.36±1.89	**93.15±1.41**
AA (%)	68.50±1.99	71.65±2.86	73.17±1.87	79.98±2.64	83.88±2.13	85.52±2.36	85.61±2.76	89.92±1.87	83.03±1.59	86.12±2.51	77.31±2.32	84.78±1.53	67.40±1.79	75.82±2.69	85.73±1.74	**91.18±1.29**
K (×100)	64.62±3.02	70.68±3.53	71.42±2.44	76.16±2.87	83.85±2.93	87.77±3.26	85.53±2.91	88.82±2.08	82.98±2.33	89.27±2.18	73.45±2.67	83.49±2.31	64.92±2.59	78.75±3.74	87.31±2.05	**91.05±1.65**

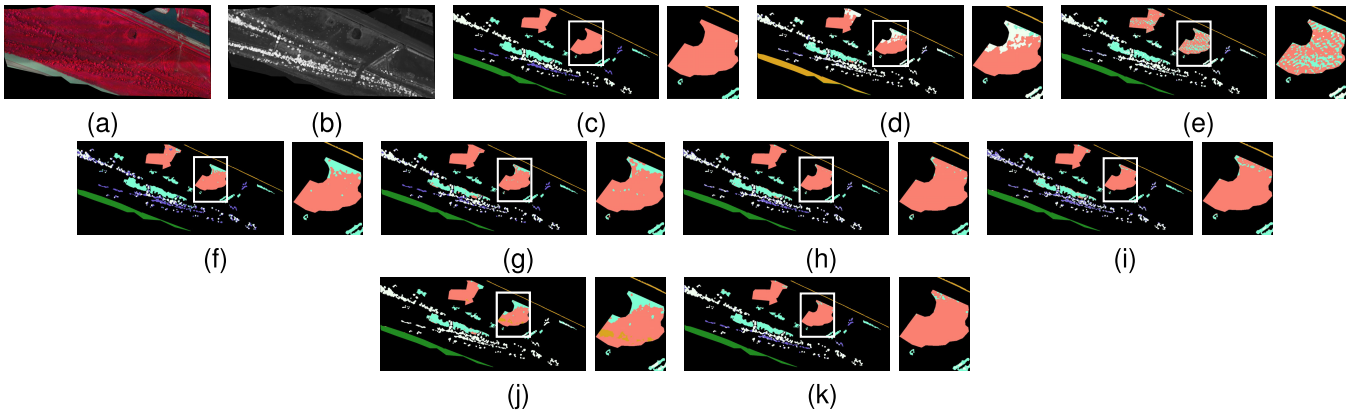


Fig. 7. Dataset visualization and classification maps of the NW-1-NW-2 task. (a) False-color image for HSIs. (b) LiDAR raster data. (c) Ground-truth map. (d)–(k) Classification maps of MFT, HCT, MCD, CLDA, TSTnet, SDEnet, LLURnet, and IDGnet, respectively.

TABLE X

CALCULATION AND SPACE COMPLEXITY OF DIFFERENT METHODS ON THE TAMARISK DATASET

Method	Parameters (NW-1-NW-2)	Time (NW-1-NW-2)		Time (NW-1-NW-3)		Time (Tamarisk-S-Tamarisk-AUT)	
		Training (s)	Testing (s)	Training (s)	Testing (s)	Training (s)	Testing (s)
MFT	298.086K	31.644	15.125	21.332	10.676	47.660	10.769
HCT	871.622K	14.949	12.901	17.877	8.426	15.248	11.731
MCD	199.652K	80.825	314.350	61.441	130.557	73.036	154.630
CLDA	199.652K	1456.822	304.928	609.160	122.238	675.645	159.825
TSTnet	5.350M	846.092	52.430	517.448	27.937	725.113	33.406
SDEnet	2.108M	438.049	31.211	292.957	19.891	400.546	22.096
LLURnet	481.902K	448.747	35.545	300.076	20.501	408.232	21.826
IDGnet	1.555M	113.214	18.972	78.879	11.285	91.231	11.941

Natural Willow datasets consists of tall arboreal species, which are more sensitive to elevation information. Methods also show different performances for cross-scene wetland classification. Taking the NW-1-NW-2 task as an example, the best OA values of MFT and HCT are 64.04% and 81.65%, respectively, which are notably lower than those of other methods. This is because, due to the traditional classification method, for MFT and HCT, domain shifts degrade cross-scene classification performance. In contrast, DA- and DG-based methods perform well in cross-scene wetland classification. In Table VIII, the OA values of MCD, CLDA, and TSTnet are 95.27%, 91.49%, and 95.69%, respectively, which are higher than those of MFT and HCT, and DG-based methods perform similarly. However, SDEnet and LLURnet show lower classification accuracy than DA-based methods. This can be attributed to their use of TD data, which is a natural advantage. Nevertheless, the proposed IDGnet obtains better classification performance than DA-based methods. As shown in Table IX, the OA of IDGnet shows improvements of 15.4%, 11.5%, 2.46%, 1.7%, 1.37%, 5.89%, and 9.48% over MFT, HCT, MCD, CLDA, TSTnet, SDEnet, and LLURnet, respectively. It can be inferred that the domain expansion of IDGnet effectively simulates domain shift in cross-scene wetland classification, and IDGnet extracts stable domain-invariant and class-specific information for wetland classification. Figs. 7–9 show visual maps for qualitative evaluation of all methods in three DG tasks. Compared with

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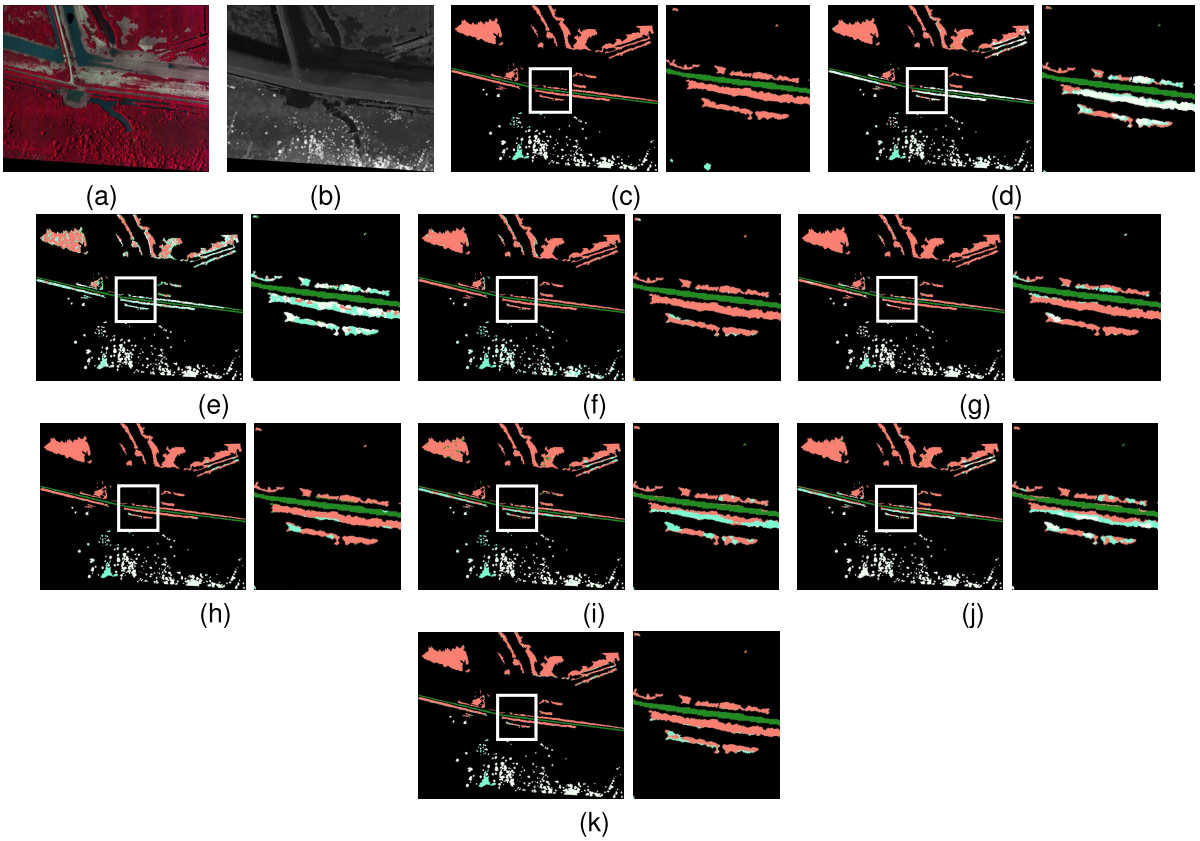


Fig. 8. Dataset visualization and classification maps of the NW-1-NW-3 task. (a) False-color image for HSIs. (b) LiDAR raster data. (c) Ground-truth map. (d)–(k) Classification maps of MFT, HCT, MCD, CLDA, TSTnet, SDEnet, LLURnet, and IDGnet, respectively.

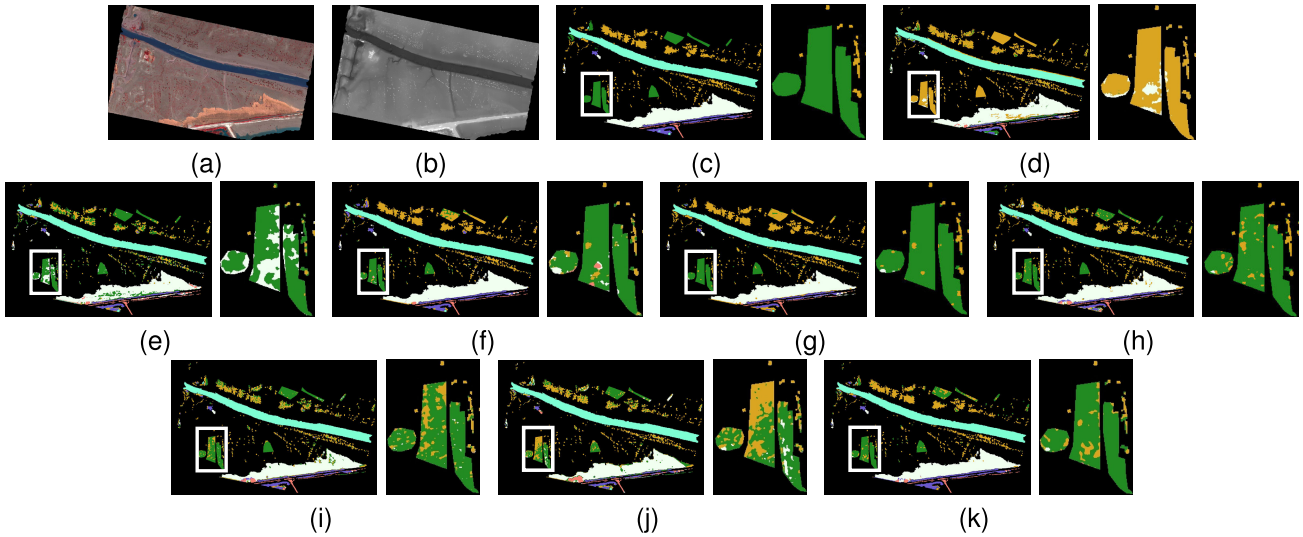


Fig. 9. Dataset visualization and classification maps of the Tamarisk-S-Tamarisk-AUT task. (a) False-color image for HSIs. (b) LiDAR raster data. (c) Ground-truth map. (d)–(k) Classification maps of MFT, HCT, MCD, CLDA, TSTnet, SDEnet, LLURnet, and IDGnet, respectively.

other methods, IDGnet provides smoother and noise-free visualization results, with a low likelihood of false detection.

E. Complexity Analysis

Model complexity is an important indicator to measure model performance. We applied the number of parameters and time cost to evaluate space and time complexity. Taking the NW-1-NW-2 task as an example, Table X reports the number of parameters in different comparison methods. It can be observed that IDGnet has the third-most parameters, which is less than TSTnet and SDEnet, which obtain the best classification accuracy in DA- and DG-based methods, respectively. IDGnet outperforms TSTnet and SDEnet with fewer parameters, which shows its efficiency. Table X also lists the training–testing time costs of different methods in three DG tasks. IDGnet achieves the highest accuracy with low time consumption. Only HCT takes less time than IDGnet, but

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as a traditional classification method, HCT is not suitable for cross-scene wetland classification. IDGnet has the best classification performance, with satisfactory model complexity, which augurs well for its rapid deployment in wetlands.

IV. Conclusion

In this work, we proposed IDGnet for cross-scene wetland classification with HSIs and LiDAR. To our knowledge, the present study was the first attempt to complete multisource remote sensing cross-scene wetland classification based on DG. Random background and semantic shifts were applied in the frequency domain for HSI and LiDAR data, respectively. This was appropriate for the simulation of domain shifts caused by the wetland environment or sensor characteristics. A strategy of multiple condition constraints reduced interdomain and intraclass differences and increased interclass differences, which was beneficial to obtaining domain-invariant and class-specific information, thus improving the generalization ability of multisource features. Experiments conducted on two wetland datasets of the Yellow River Delta revealed the effectiveness of IDGnet, which could outperform state-of-the-art models in the cross-scene wetland classification task. It should be noted that this work did not discuss cross-scene wetland classification in different regions and different seasons, which was limited by the inability to obtain data. We hope to address this problem in our future work.

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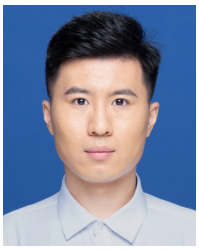
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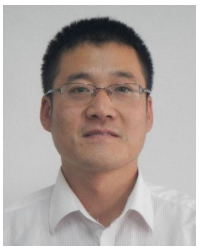
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